Scavenging

At the end of exhaust stroke the exhaust valve is open and at the start of intake stroke, the inlet valve is kept open. At this time, fresh air – fuel mixture for the next cycle will pushes out the exhaust gases of the previous cycle. Thus, Scavenging is the process of removal of exhaust gases by blowing in fresh air.

Valve Over Lap

At the end of exhaust stroke the exhaust valve is open and at the start of intake stroke, the inlet valve is kept open. The time duration in which both the valves are kept open is called as valve overlap.

Advantages

- ❒ Fuel economy
- ❒ Lubrication oil Consumption is less
- ❒ Can be used in different vehicle
- ❒ Thermal Efficiency is high
- ❒ Volumetric Efficiency is high
- ❒ Low wear and tear

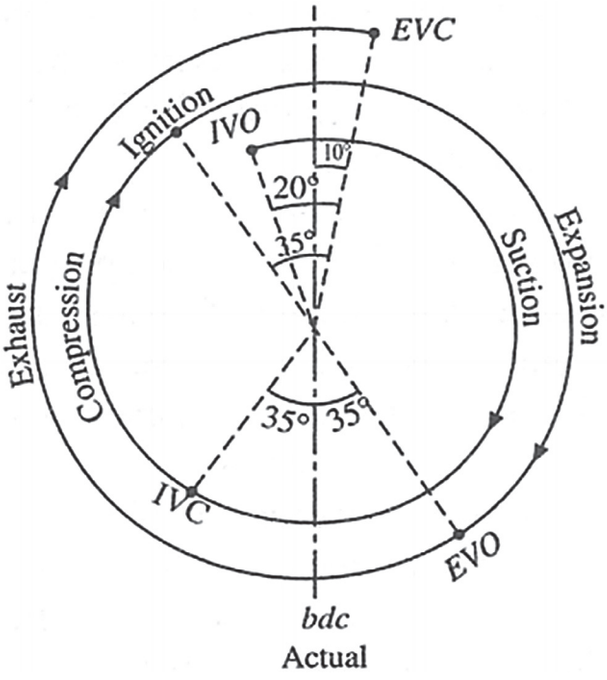
Disadvantages

- ❒ Number of parts are more
- ❒ Mechanical Efficiency is low
- ❒ Maintenance cost is more
- ❒ Complicated design
- ❒ More space is required

e) Valve Timing Diagram For A Four Stroke Petrol Engine

In a four stroke engine, an engine cycle to produce power is done on every two revolution of crankshaft. Theoretically it may be assumed that the valves open and close and the spark (or injection of fuel)occurs at the engine dead centres. However, in actual operation, the valves do not operate at dead centre positions but operate some degree on either side of the dead centres. The opening occurs earlier and the exhaust continues even at later crank angles. The ignition is also timed to occur in advance of the completion of compression stroke.

The timings of this sequence of events can be shown graphically in terms of crank angles from dead centre position. This diagram is known as valve timing diagram. It is shown in figure 5.4(e).



TDC = Top dead centre, BDC = Bottom dead centre, IVO = Inlet valve opens IVC = Inlet valve closes, EVO = Exhaust valve opens, EVC = Exhaust valve closes

Figure 5.4(e) Valve Timing Diagram For Four Stroke Petrol Engine

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